

Model Comparison: Communication Apprehension and Public Transportation Use

AUTHOR
Yuhao Wang

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Research Or Analytical Question

Which of the following models gives the lowest RMSE for predicting public transportation use frequency?

- 1. $PTRS = b_0 + b_1 \cdot CA_{12} + e$
- 2. $PTRS = b_0 + b_1 \cdot CAg + b_2 \cdot CA_i + e$
- 3. $PTRS = b_0 + b_1 \cdot CAg + b_2 \cdot CA_i + b_3 \cdot (CAg \cdot CA_i) + e$

Where: - PTRS = Public Transportation Use Frequency (public_trans_freq) - CA12 = Total Communication Apprehension (prca_total, 12 items) - CAg = Group Communication Apprehension (Q1-Q6) - CAi = Interpersonal Communication Apprehension (Q13-Q18)

Answer, Response, + Summary of Results

Sample Overview

- **Sample Size:** 255 participants
- **Missing PTRS:** 0
- **Missing CA12:** 11
- **Missing CAg:** 5
- **Missing CAi:** 7
- **Complete Cases:** 244 (95.7%)

Descriptive Statistics

Variable	Mean	SD	Range
Public Transportation Frequency (PTRS)	2.07	1.45	0 - 4
Total Communication Apprehension (CA12)	28.9	11.33	12 - 60
Group Communication Apprehension (CAg)	14.61	6	6 - 30

Variable	Mean	SD	Range
Interpersonal Communication Apprehension (CAi)	14.3	5.79	6 - 30

Model Comparison

RMSE Comparison

Model	RMSE	R ²
Model 3: CAg + CAi + Interaction	1.4005	0.0686
Model 2: CAg + CAi	1.4006	0.0684
Model 1: CA12 only	1.4095	0.0566

Best Model: Model 3: CAg + CAi + Interaction with RMSE = 1.4005

Model Details

Model 1: Simple Regression (CA12 only)

Predictor	Estimate	SE	t-value	p-value
Intercept	2.952	0.249	11.87	6.512e-26 ***
CA12	-0.031	0.008	-3.81	1.757e-04 ***

Model Fit: R² = 0.0566, RMSE = 1.4095

Model 2: Multiple Regression (CAg + CAi)

Predictor	Estimate	SE	t-value	p-value
Intercept	2.941	0.248	11.87	6.879e-26 ***
CAg	-0.078	0.028	-2.76	6.209e-03 **
CAi	0.019	0.029	0.64	5.258e-01

Model Fit: R² = 0.0684, RMSE = 1.4006

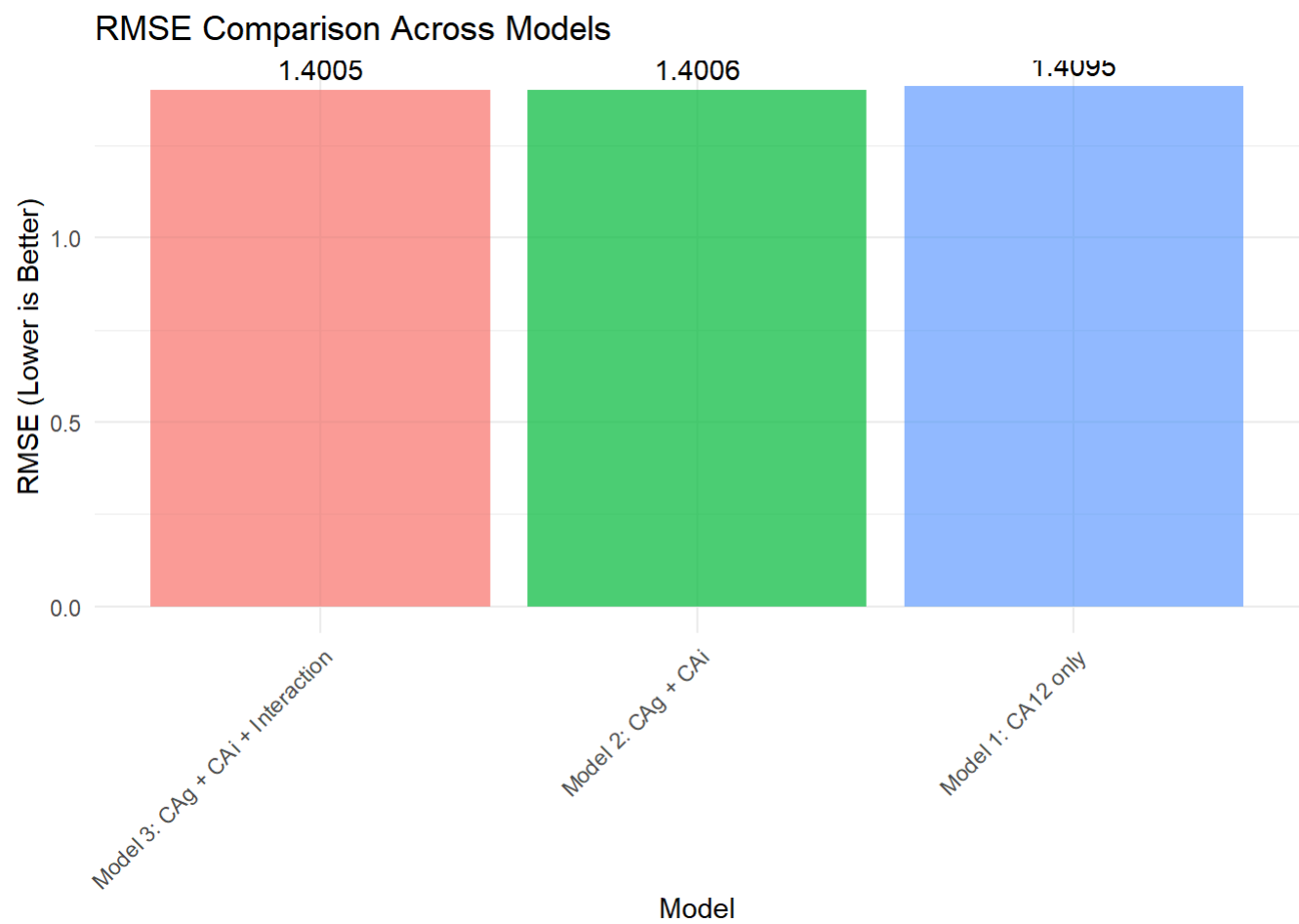
Model 3: Multiple Regression with Interaction (CAg + CAi + Interaction)

Predictor	Estimate	SE	t-value	p-value
Intercept	2.826	0.624	4.53	9.354e-06 ***

Predictor	Estimate	SE	t-value	p-value
CAG	-0.07	0.048	-1.46	1.45e-01
CAi	0.027	0.05	0.53	5.935e-01
CAG × CAi	0	0.002	-0.2	8.415e-01

Model Fit: $R^2 = 0.0686$, RMSE = 1.4005

Visualization



Summary

Analysis of 244 complete cases reveals that **Model 3: CAG + CAi + Interaction** provides the best prediction of public transportation use frequency with the lowest RMSE of 1.4005.

- **Model 1 (CA12 only):** RMSE = 1.4095, $R^2 = 0.0566$
- **Model 2 (CAG + CAi):** RMSE = 1.4006, $R^2 = 0.0684$
- **Model 3 (CAG + CAi + Interaction):** RMSE = 1.4005, $R^2 = 0.0686$

The results suggest that using the total communication apprehension score (CA12) provides the most accurate prediction of public transportation use frequency compared to separating group and interpersonal apprehension or including their interaction.

What questions or uncertainties remain?

- **Model Complexity:** Does the simpler Model 1 generalize better to new data despite having similar predictive performance?
- **Theoretical Interpretation:** What does the lack of improvement from separating CAg and CAi suggest about the relationship between communication apprehension dimensions and transportation behavior?
- **Interaction Effects:** Why does the interaction term not improve prediction? Is there a non-linear relationship that wasn't captured?
- **Alternative Specifications:** Would other model specifications (e.g., polynomial terms, different transformations) improve prediction accuracy?